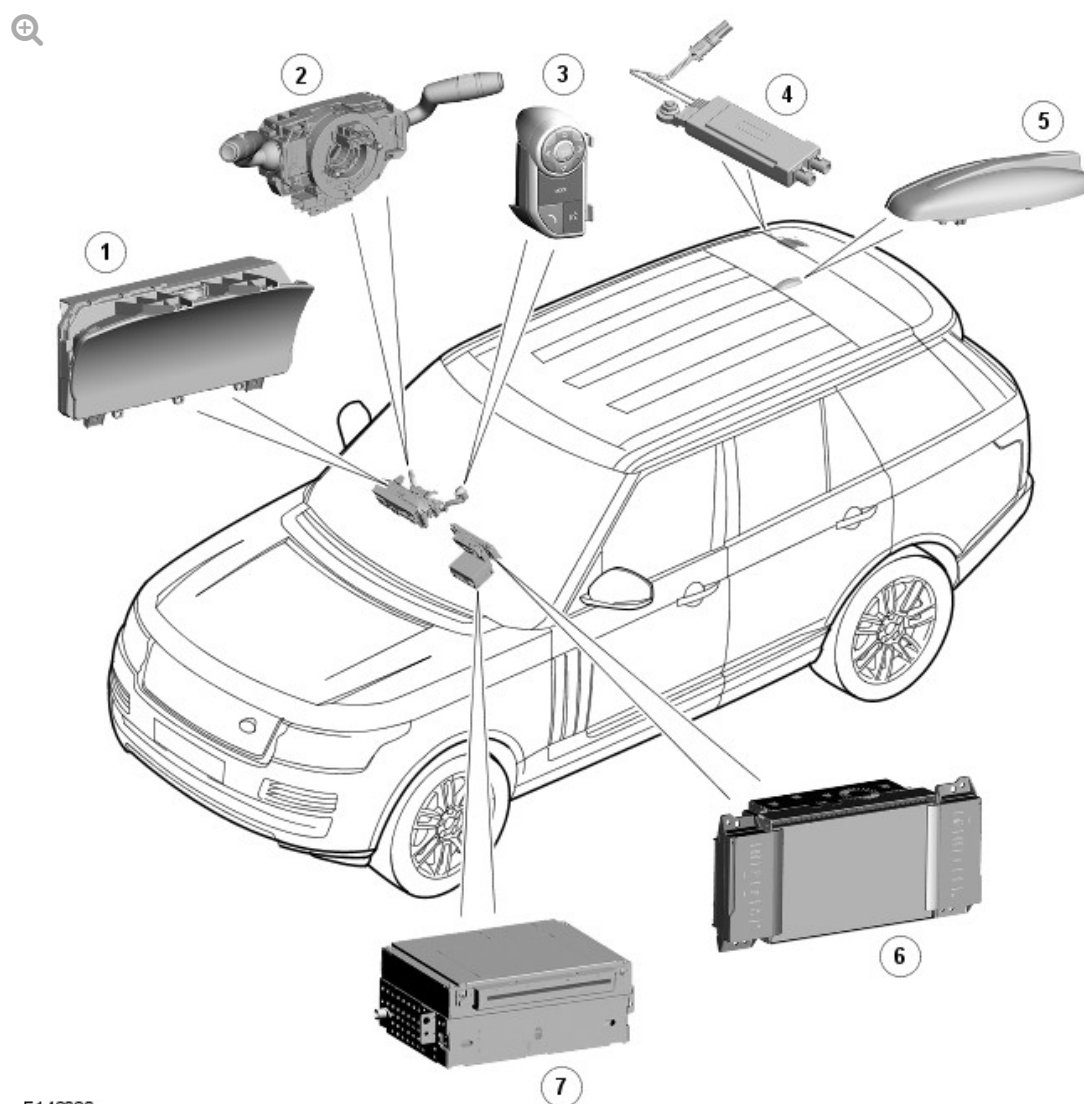


2016.0 RANGE ROVER (LG), 415-01

INFORMATION AND ENTERTAINMENT SYSTEM

DESCRIPTION AND OPERATION

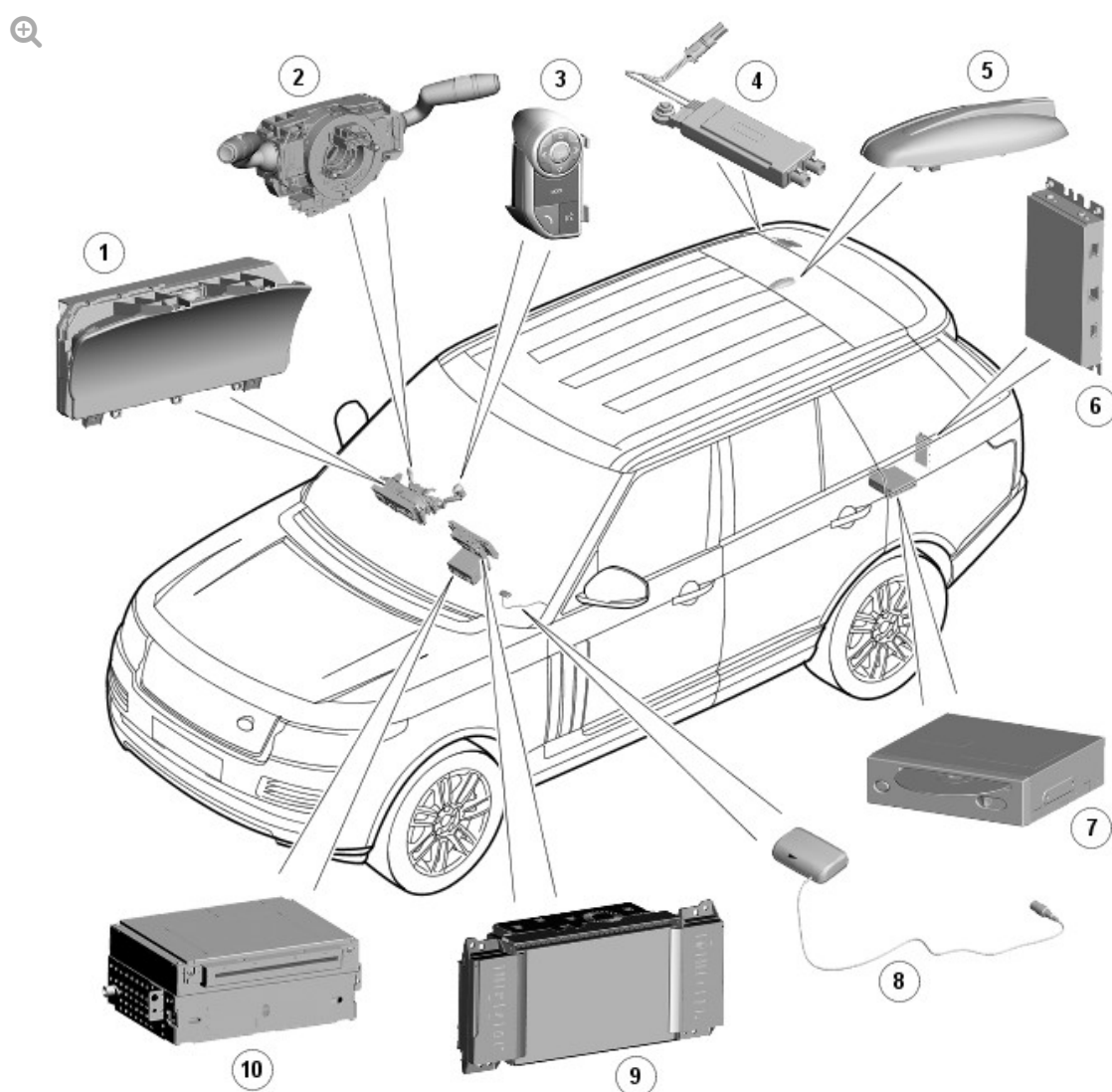
COMPONENT LOCATION - ROW



E142020

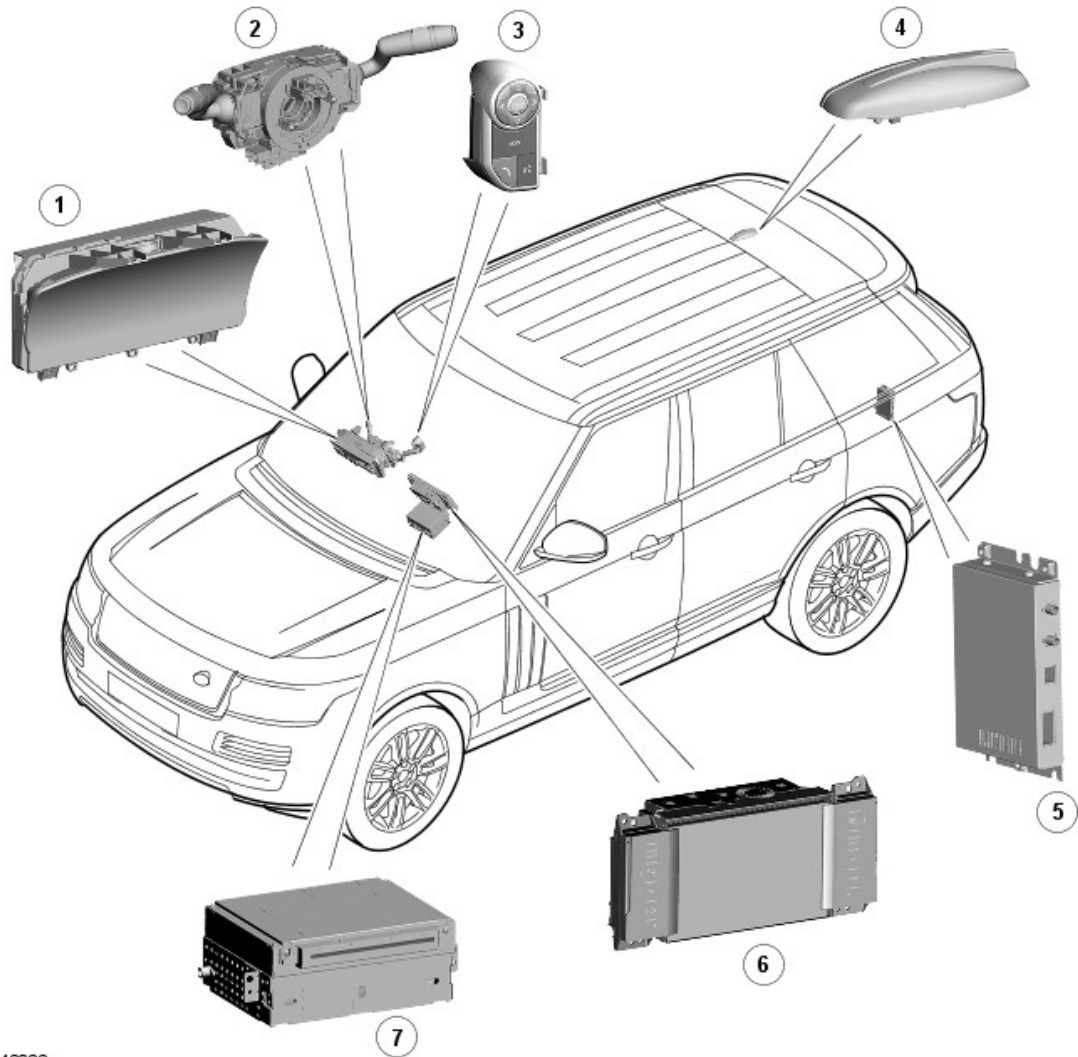
ITEM	DESCRIPTION
1	Instrument cluster
2	Clockspring
3	Left steering wheel switchpack
4	Antenna amplifier FM2/digital radio/TMC
5	Roof pod
6	Touch Screen (TS)
7	Integrated Audio Module (IAM)

COMPONENT LOCATION - JAPAN



ITEM	DESCRIPTION
1	Instrument cluster
2	Clockspring
3	Left steering wheel switchpack
4	Antenna amplifier - FM2/Digital Radio/VICS
5	Roof pod
6	Video interface module
7	Navigation control module
8	VICS beacon antenna
9	Touch Screen (TS)
10	Integrated Audio Module (IAM)

COMPONENT LOCATION - ASIA



E142022

ITEM	DESCRIPTION
1	Instrument cluster
2	Clockspring
3	Left steering wheel switchpack
4	Roof pod
5	Navigation control module
6	Touch Screen (TS)
7	Integrated Audio Module (IAM)

OVERVIEW

The navigation system provides audible and visual route guidance information to enable the driver to reach a desired destination. The system allows the driver to choose the route using minor or major roads or highways with the option of three routes. The system provides the driver with the option of on-road and off-road navigation.

The navigation system is integrated with the audio/video system and shares a number of components common to all systems. Map information is stored on a Hard Disc Drive (HDD) located in the Integrated Audio Module (IAM). Map uploads to the hard drive can be uploaded by the customer from a USB (universal serial bus) memory stick (not applicable to Japan/Asia specification vehicles).

The navigation system has various levels of user control through the Touch Screen (TS) and the Land Rover Voice system. System volume adjustment can be made using the Integrated Control Panel (ICP), TS and left steering wheel switchpack.

There are a number of navigation system variants specific to various markets. On all systems the GPS (global positioning system) signal is received by the GPS antenna located in the roof pod.

The European navigation system includes the Traffic Messaging Channel (TMC) function, which receives traffic information from an FM (frequency modulation) antenna integrated into the rear spoiler. On a pre-selected route, the system will offer re-routing options depending on traffic conditions.

All North America Specification (NAS) vehicles are configured to receive TMC. TMC is transmitted in the USA and is available in areas of other NAS markets.

The navigation system is primarily controlled from the TS which is located in the center of the instrument panel. Control signals from the TS are sent on the Media Oriented System Transport (MOST) ring to the navigation software within the IAM. The IAM uses a dedicated Gigabit Video InterFace

(GVIF) bus to transmit video signals to the TS.

The navigation system uses the following components:

- Integrated Audio Module (IAM)
- Single or Dual-view Touch Screen (TS)
- Roof pod GPS antenna
- Instrument cluster.

The dual-view TS allows the front seat passenger to view television and video images when the car is being driven. The dual-view screen allows the driver to see the navigation or other system screens but not the TV or video when the vehicle is moving. The screen can be switched between single and dual view using a button on the Touch Screen (TS).

NOTE:

Due to legislation, the NAS markets do not receive this dual-view option.

Japanese market vehicles have a modified system from other markets. These vehicles have an additional DVD (digital versatile disc) navigation computer module and a navigation video interface module located in the luggage compartment.

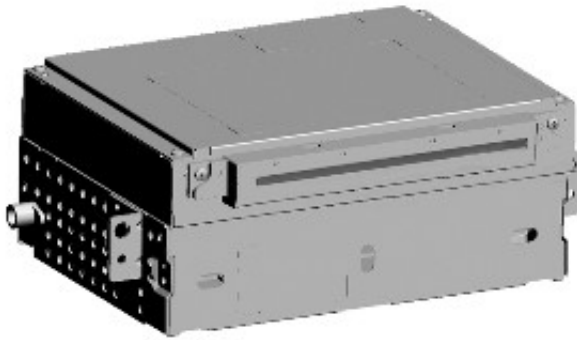
The Japanese navigation system includes the Vehicle Information and Communication System (VICS) function. The VICS supplies information to enable the navigation computer to re-route the navigation guidance or to inform the vehicle driver of traffic conditions in the vehicles vicinity. VICS information is provided to the system through an FM antenna integrated into the rear spoiler and a VICS beacon located on the left side of the instrument panel.

Asia markets have a unique system which is fitted to the vehicle by the

dealer or port of entry. These vehicles have a separate navigation computer module located in the luggage compartment.

DESCRIPTION - ROW

INTEGRATED AUDIO MODULE (IAM)



E141841

NOTE:

The Japan/Asia satellite navigation system does not store map data on the IAM. All other functions of the IAM are applicable to the Japanese market. Refer to the following sections 'Description - Japan' or 'Description - Asia' for details of the Japan/Asia navigation system.

The IAM is located in a central position in the instrument panel, behind the Integrated Control Panel (ICP).

Three levels of IAM are available; IAM 1, IAM 2 and IAM3. Only the IAM 3 module supports navigation.

The IAM is connected on the MOST ring to the other audio system components. The driver can control navigation functions by using soft keys on the Touch Screen (TS), left steering wheel switchpack or by voice commands.

A 40 GB hard drive is used for storing the information for satellite

navigation and music files. A 10GB partition is provided for storing music files, the remaining 30GB is used for map data storage.

The integral hard drive for the navigation system removes the requirement of a separate navigation computer module usually found in the rear luggage compartment. The IAM stores the navigation map data locally within the 30GB hard drive partition. By storing the information in this way and processing it within the IAM, navigation display, route calculation speeds and accuracy are vastly improved. Map upgrades and software now have to be loaded directly into the IAM from a DVD or USB memory device.

The map images are transmitted from the IAM to the TS via a Low Voltage Differential Signal (LVDS) link cable. Turn by turn instructions are also available, these are displayed in the instrument cluster via a medium speed comfort CAN (controller area network) link between the instrument cluster and TS.

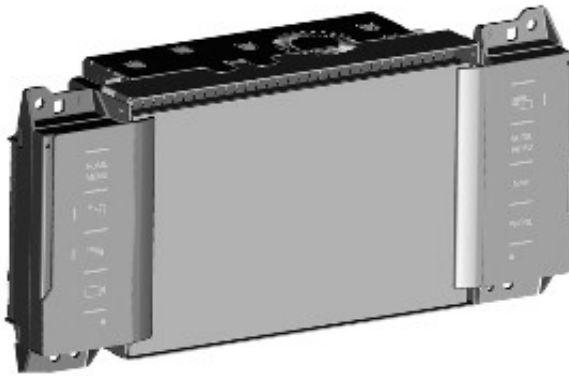
The IAM communicates on the MOST ring with the rest of the audio system. If the IAM is replaced it must be configured as a new module using an approved Land Rover diagnostic system.

Calibration of the IAM using an approved Land Rover diagnostic system enables updates to be downloaded as new technology becomes available or any fault concerns require software updates.

For further details on IAM functionality refer to the Audio system.

For additional information, refer to: [Audio System](#) (415-01 Information and Entertainment System, Description and Operation).

TOUCH SCREEN (TS)



E141842

The Touch Screen (TS) is located in the center of the instrument panel. The TS comprises an 8 inch color, touch sensitive display, with two separate touch sensitive TS switchpacks located on either side.

Two versions of the TS are available; single view and dual view. Additionally TS switchpacks differ depending on system specification; for example, if Park Assist, navigation, dual view or rear view camera is specified on the vehicle, these soft keys will replace Setup, Audio Settings, Mode and display on/off respectively. The functions can still be accessed via the home menu using the touch screen on the TS.

The TS is the Bus Master for the Media Oriented System Transport (MOST) system and contains the timing master for the MOST system.

The TS is the primary user interface of the audio system. It communicates with the rest of the audio/infotainment system on the MOST ring and allows control of the audio system and other infotainment systems from a single point. No configuration procedure is required if the TS or TS switchpacks are replaced.

The TS switchpacks comprise a number of touch sensitive soft keys. The soft keys are not visible when the system is powered down. When the system is active the soft keys are illuminated. When a soft key is pressed, a line above and below the soft key is illuminated to indicate the press has been accepted.

The driver's view is event driven, for example. if reverse gear is selected the

rear view camera is displayed automatically, overriding the displayed navigation or other information.

For further details on TS functionality refer to the Audio system.

For additional information, refer to: [Audio System](#) (415-01 Information and Entertainment System, Description and Operation).

ROOF POD



E141520

The navigation GPS antenna is located in the roof pod. Depending on vehicle specification the roof pod can contain a GPS, digital radio or satellite radio antennas.

The roof pod is fitted externally on the roof panel and is attached to the roof panel with two flanged nuts which screw onto captive studs in the roof pod. Two co-axial connectors provide for the connection for the GPS antenna and the satellite or digital radio antenna.

TRAFFIC MESSAGE CHANNEL (TMC) ANTENNA

Data messages for TMC are received through the FM antenna and antenna amplifier located in the rear spoiler. The antenna is also used by the IAM for reception of FM2 and Digital Radio transmissions.

For further details on the antenna systems refer to the Antenna system.

For additional information, refer to: [Antenna](#) (415-01 Information and Entertainment System, Description and Operation).

ADVANCED VOICE CONTROL SYSTEM

NOTE:

Voice control of the navigation system is not available in Japan or Asia markets.

The advanced voice system provides the driver with the option of voice control for a range of supported functions. The following systems include Voice functionality:

- Radio, satellite radio and Digital Radio (if fitted)
- CD player
- Hard disk drive/virtual CD player
- USB and auxiliary connections
- Navigation system
- Bluetooth® Telephone system
- Vehicle notepad.

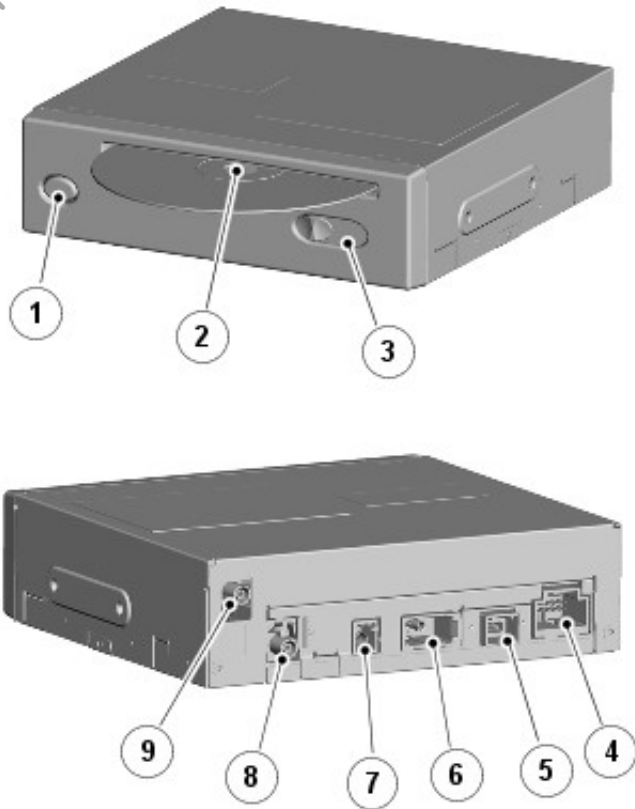
For further details of the advanced voice control system refer to the Voice Control section.

For additional information, refer to: [Antenna](#) (415-01 Information and Entertainment System, Description and Operation).

DESCRIPTION - JAPAN

The Japan satellite navigation system uses a separate navigation computer module with map data supported by an DVD disk. Additional components are: a navigation computer module and a navigation video interface module.

NAVIGATION COMPUTER MODULE



E 142023

ITEM	DESCRIPTION
1	DVD eject button
2	DVD loading slot
3	DVD eject lock
4	Power and ground connections
5	GVIF video output connector to video interface module
6	MOST connector
7	VICS beacon antenna connector
8	GPS (global positioning system) antenna connector
9	VICS FM antenna connector

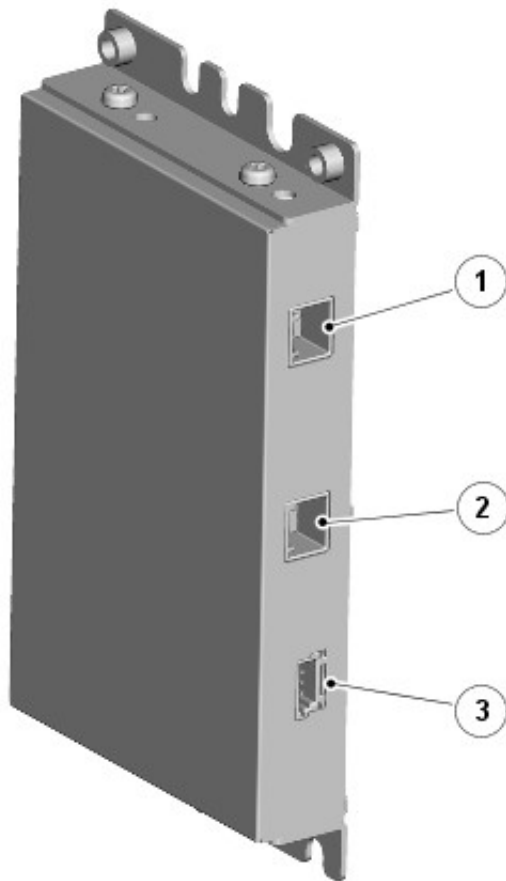
A navigation computer module is located in the left rear of the luggage compartment, The module is located on a bracket secured to the luggage compartment floor and is accessible via an opening in the spare wheel well

carpet.

The navigation computer module is a DVD drive which reads map data direct from a DVD. The navigation computer module is connected on the MOST ring and communicates with the TS to initiate navigation video and audio output. The GPS antenna is connected directly to the navigation computer module.

The navigation computer module outputs the video signals in a Gigabyte Video InterFace (GVIF) format to a navigation video interface module which converts the GVIF input to a Low-Voltage Differential Signalling (LVDS) video signal output which is then passed to the TS. Audio output is on the MOST ring to the audio amplifier module. VICS FM transmission signals are received by the navigation computer module via an FM antenna and a VICS antenna amplifier. Infrared and RF microwave VICS transmissions are also received by the VICS beacon antenna, located on the top of the instrument panel, and are passed to the navigation computer module.

VIDEO INTERFACE MODULE



E 142024

ITEM	DESCRIPTION
1	Power, ground and 5V signal voltage from TS connector
2	LVDS video output to TS connector
3	GVIF video input from navigation control module connector

A navigation video interface module is located in the left rear of the luggage compartment, attached to a bracket shared with the TV control module.

The video interface module is required to convert the GVIF video output to LVDS video signal which is compatible with the TS.

A 5V signal output from the TS is connected to the video interface module. The signal voltage initiates a power up of the video interface module when the TS is active.

VEHICLE INFORMATION AND COMMUNICATION SYSTEM (VICS)

The VICS is a similar system to the TMC used outside of Japan. VICS is a unique system to Japan and provides countrywide coverage and broadcasts of real-time traffic and weather information.

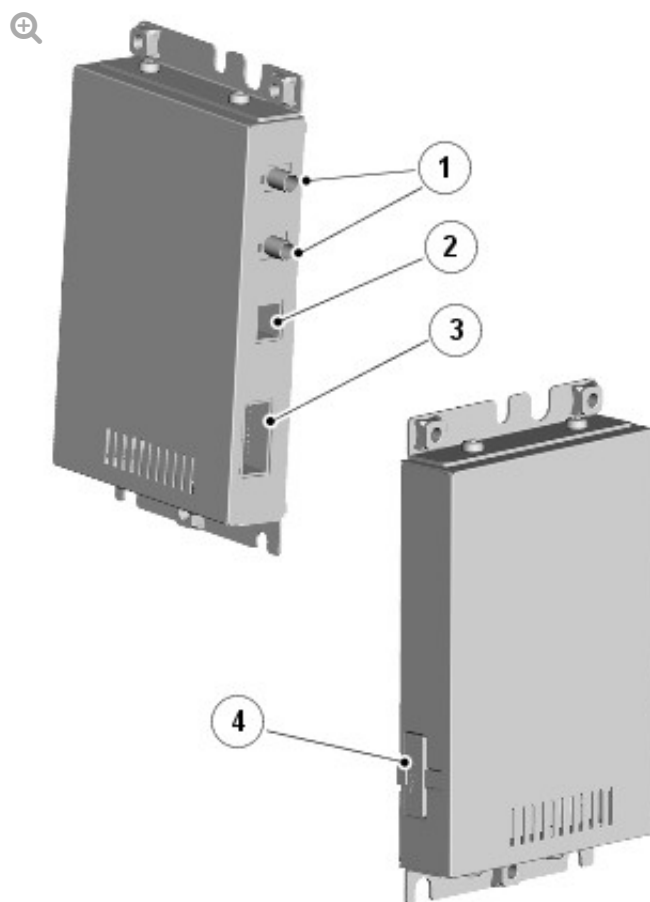
The VICS has three methods of transmitting the traffic data to the vehicle's navigation system: FM, microwave RF and infrared.

The microwave and infrared transmissions are both received by the VICS beacon antenna located on the instrument panel.

The same information is also transmitted on a FM wavelength and is received by the FM antenna located in the rear spoiler and uses the audio system antenna amplifiers to pass the signals to the navigation computer module.

DESCRIPTION - ASIA

NAVIGATION COMPUTER MODULE



E 142025

ITEM	DESCRIPTION
1	GPS antenna connector
2	LVDS video output to TS
3	Power, CAN and audio connector
4	SD storage card

In some Asia markets, a unique navigation computer module is fitted. In some cases this can be 'factory' fitted or fitted at PDI (pre-delivery inspection) by the dealer or at Port of Entry (POE). The medium speed comfort CAN based navigation computer module is located in the left rear of the luggage compartment, attached to a bracket shared with the TV control module.

The navigation computer module outputs the video signals in a LVDS format direct to the TS. Audio output is passed to the IAM which converts the

signals and passes them to the audio amplifier module on the MOST ring. When Audio is required, such as a Voice guidance instruction, the Asia navigation computer module communicates to the vehicle audio system using a hard wire connection between the TS and the Asia navigation computer module. Touch screen co-ordinates and vehicle power mode status is obtained through the medium speed comfort CAN. Map data is stored via a multimedia Secure Digital (SD) card accessible through an access point on the module.

A 5V signal output from the TS is connected to the navigation control module. The signal voltage initiates a power up of the navigation control module when the TS is active.

OPERATION

The navigation system receives GPS information via the GPS antenna. The GPS signals are used by the navigation system to calculate the vehicle's position. Once the driver has input a desired destination, the navigation system can calculate a route, based on the driver's pre-determined preferences or the default settings in the navigation system

The navigation system is accessed from the Touch Screen (TS) home menu.

Navigation is initiated by the driver inputting a destination. This can be achieved via the TS by:

- Entering an address
- Entering a post code
- Choosing a previous destination
- Choosing a point of interest from the map disc database
- Choosing the home location
- Choosing a memory stored location.

The driver is then guided to the destination by a scrolling map display and

voice guidance. The display can be varied by scale and display type.

Selection of 'Navigation' on the TS home menu and subsequent sub-menu selection sends a control request signal to the IAM for Rest Of World (ROW) markets or navigation computer module (Japan only) on the Media Oriented Systems Transport (MOST) ring. In Asia markets the TS control request is passed to the navigation computer via the medium speed comfort CAN bus. The requested control information is processed by the Integrated Audio Module (IAM) for ROW or navigation computer module for Japan and Asia only.

On ROW vehicles, if voice guidance is operational, the voice signals are passed from the IAM to the audio amplifier module on the MOST ring for output on the speaker system.

On Japan vehicles, if voice guidance is operational, the voice signal information is relayed from the navigation computer module on the MOST to the audio amplifier module for output on the speaker system.

On Asia vehicles, if voice guidance is operational, the voice signal information is relayed from the navigation computer module, via an analogue audio signal to the IAM. The IAM converts the analogue signals which are then passed to the audio amplifier module on the MOST ring for output on the speaker system.

The navigation audio output is through the front speakers whilst the background audio, for example radio or CD (compact disc), is played at a reduced volume on the rear speakers. On Asia systems the radio or CD output is paused while navigation audio output is operating.

The GPS signal is available to the navigation system at all times when the vehicle ignition is switched on.

Navigation user voice commands are made using the Land Rover Advanced Voice system. The TS processes the analogue signal from the Voice switch on the left steering wheel switchpack. This is passed from the TS onto the

MOST system to the Land Rover Advanced Voice control software which is integral with the IAM (ROW) or navigation computer (Japan). Voice control of the navigation system is not available on Asia specification vehicles.

The TS sends an instruction via the MOST ring to the IAM to turn on the microphone facility. The microphone is hardwired to the IAM. For the ROW system, spoken voice commands are processed by the IAM. The processed commands are then sent to the TS to determine which control signals need to be sent to the navigation system.

For the Japan system, the analogue voice signals are relayed from the IAM via the MOST ring directly to the navigation computer module (for processing). Navigation commands are handled internally within the navigation computer module without the need for communicating with the TS for control.

Traffic data from the TMC or the VICS is processed by the IAM or navigation computer module (Japan only), distributed to the TS with any supporting voice instruction relayed through the MOST ring to either the IAM or audio amplifier module, dependant on equipment level, for output on the speaker system.

TRAFFIC MESSAGE CHANNEL (TMC)

NOTE:

TMC is not available in all markets.

The TMC is a specific application of the FM Radio Data System (RDS) or Radio Broadcast Data System (RBDS) (NAS only) used for broadcasting real-time traffic and weather information. Data messages are received and decoded by the IAM. The IAM processes the received information and alerts the driver and offers alternative route guidance to avoid the incident.

Each traffic incident is sent as a TMC message. One message consists of an event code and a location code in addition to time details. The message is

coded and can be translated by the IAM into the market language. Location code tables assign numbers to locations on the road network. Those location tables are integrated in the maps stored on the IAM hard disk drive. The source of traffic information is typically police, traffic cameras and local network stations.

The TMC system uses the existing FM antenna located in the rear spoiler and the audio system antenna amplifiers to pass the signals to the IAM.

VEHICLE INFORMATION AND COMMUNICATION SYSTEM (VICS) - JAPAN ONLY

The VICS is a similar system to the TMC used outside of Japan. VICS is a unique system for Japan and provides countrywide coverage and broadcasts of real-time traffic and weather information. The VICS has three methods of transmitting the traffic data to the vehicle's navigation system: FM, microwave RF and infrared. The microwave and infrared transmissions are both received by the VICS beacon antenna. The same information is also transmitted on a FM wavelength and is received by the FM antenna located in the rear spoiler and uses the audio system antenna amplifiers to pass the signals to the navigation computer module.

The VICS supplies information to enable the navigation computer module to re-route the navigation guidance or to inform the vehicle driver of traffic conditions in the vehicles vicinity.

The Radio Frequency (RF) transmissions are generally transmitted from road side beacons mainly on expressways. The information transmitted is as follows:

- Traffic congestion
- Travel time to next intersection
- Traffic conditions in surrounding areas and expressway turn offs
- Traffic accidents
- Speed limits

- Lane regulations
- Tire change
- Parking availability at expressway service areas and parking areas.

Infrared transmissions are transmitted from road side beacons on major trunk roads. The information transmitted is:

- Traffic congestion and travel time
- Traffic accidents
- Breakdowns
- Road works restrictions
- Parking availability.

FM transmissions are broadcast as part of the FM multiplex broadcasting system from NHK (Japanese Broadcasting Corporation) FM stations.

Information transmitted is:

- Traffic congestion and travel time for wide areas
- Traffic accidents, road works, speed limits and lane restrictions for a wide area
- Parking availability information.

The traffic data is split from the normal FM transmissions by the FM antenna amplifier module.

OFF-ROAD NAVIGATION

NOTE:

Off-Road Navigation is not available on Asia/Japan navigation systems.

Off-road navigation is selected pressing the 'NAV' hard key on the TS then

select the Off-road nav icon.

Off-road navigation maps are similar to the on-road maps but have additional features and information, for example; a large compass display, heading, altitude, latitude and longitude.

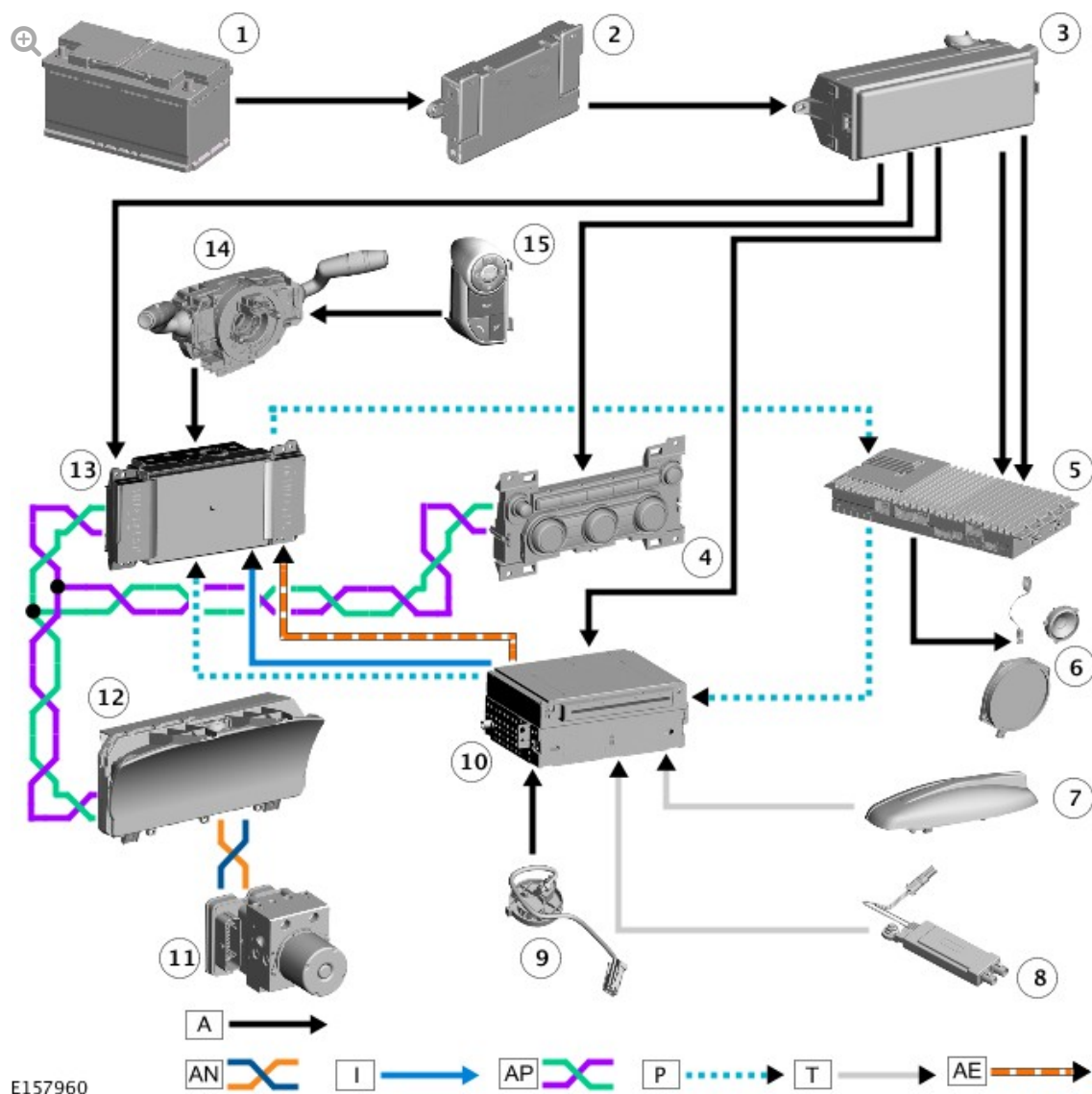
When switching to off-road navigation mode during on-road guidance, the current highlighted on-road route will be removed from the map and on-road guidance will be suspended. Way points and destination icons will remain displayed. The destination is marked with a double circle.

When exiting off-road guidance, the system recalculates the route and reverts to on-road guidance.

NOTE:

TMC/VICS is not available during off road navigation.

CONTROL DIAGRAM - ROW



ITEM

DESCRIPTION

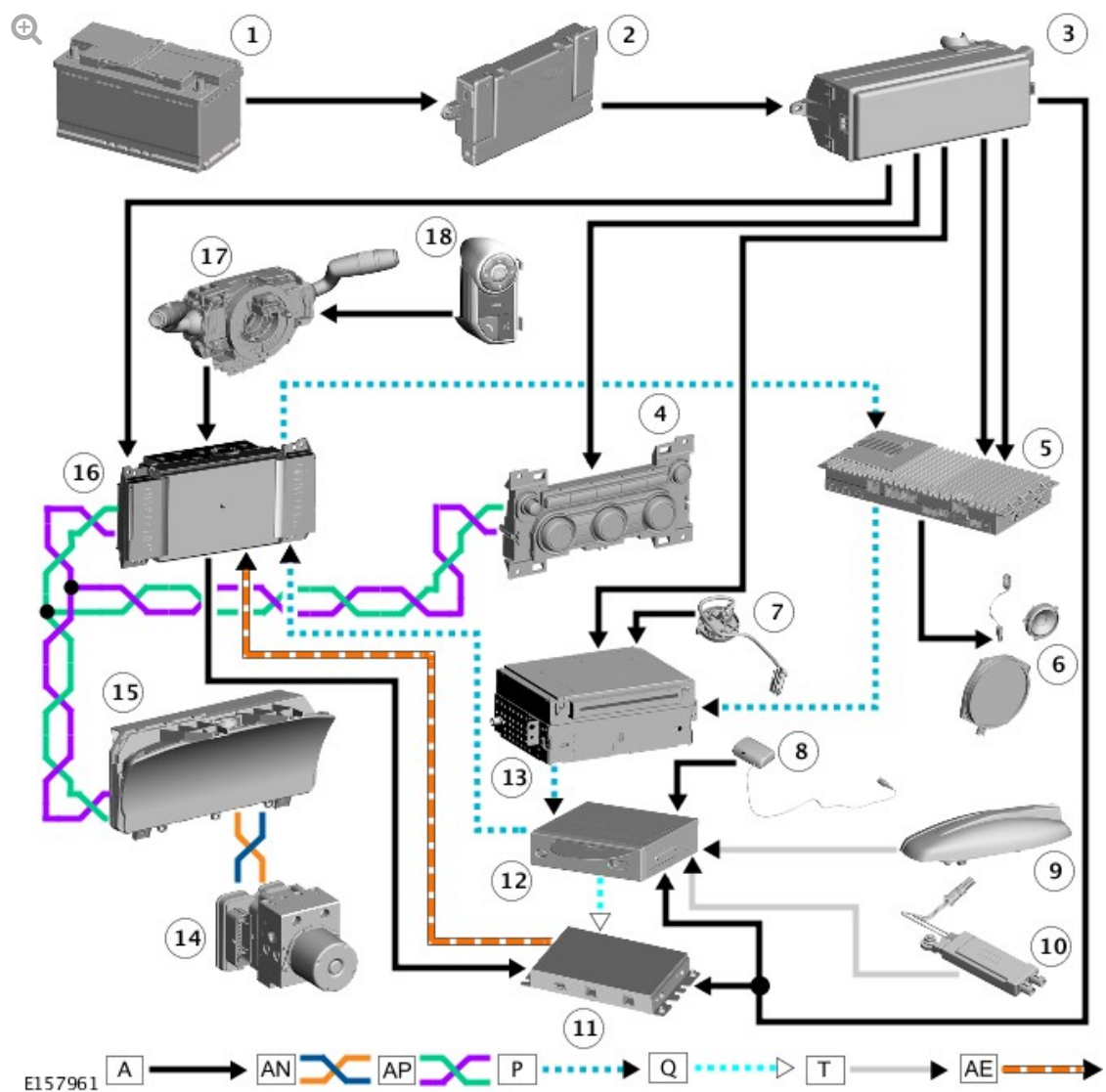
-	A = Hardwired; I = CVBS; AN = High Speed Powertrain CAN (controller area network); AP = Medium Speed Comfort CAN (controller area network) bus; P = MOST; T = Coaxial; AE = LVDS
1	Battery
2	Battery Junction Box (BJB)
3	Rear Junction Box (RJB)
4	Integrated Control Panel (ICP)
5	Audio amplifier module
6	Vehicle speakers
7	Roof pod
8	Antenna amplifier FM2/Digital Radio/TMC

9	Microphone (only if voice system fitted)
10	Integrated Audio Module (IAM)
11	Anti lock Brake System (ABS) control module
12	Instrument cluster
13	Touch Screen (TS)
14	Clockspring
15	Left steering wheel switchpack

CONTROL DIAGRAM - JAPAN

NOTE:

A = Hardwired; **AN** = Powertrain High speed CAN; **AP** = Comfort Medium speed CAN; **P** = MOST; **Q** = GVIF; **T** = Coaxial; **AE** = LVDS



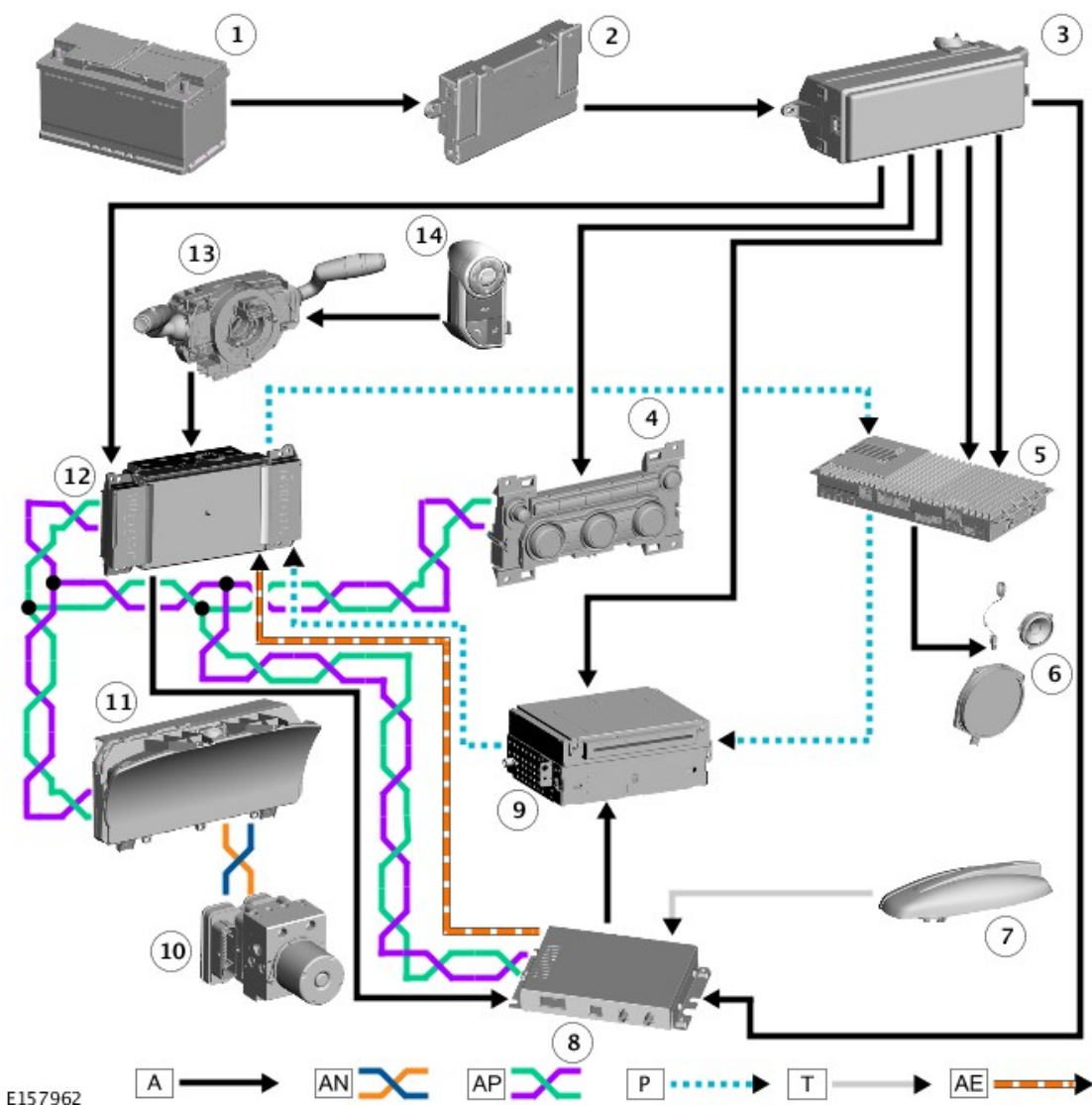
ITEM

DESCRIPTION

-	A = Hardwired; I = CVBS; AN = High Speed Powertrain CAN (controller area network); AP = Medium Speed Comfort CAN (controller area network) bus; P = MOST; Q = GVIF; T = Coaxial; AE = LVDS
1	Battery
2	Battery Junction Box (BJB)
3	Rear Junction Box (RJB)
4	Integrated Control Panel (ICP)
5	Audio amplifier module
6	Vehicle speakers
7	Microphone (only if voice system is fitted)
8	VICS beacon antenna

9	Roof pod
10	Antenna amplifier - FM2/Digital Radio/VICS
11	Video interface module
12	Navigation computer module
13	Integrated Audio Module (IAM)
14	Anti lock Brake System (ABS) control module
15	Instrument cluster
16	Touch Screen (TS)
17	Clockspring
18	Left steering wheel switchpack

CONTROL DIAGRAM - ASIA



E157962

ITEM

DESCRIPTION

-	A = Hardwired; I = CVBS; AN = High Speed Powertrain CAN (controller area network); AP = Medium Speed Comfort CAN (controller area network) bus; P = MOST; T = Coaxial; AE = LVDS
1	Battery
2	Battery Junction Box (BJB)
3	Rear Junction Box (RJB)
4	Integrated Control Panel (ICP)
5	Audio amplifier module
6	Vehicle speakers
7	Rood pod

8	Navigation control module
9	Integrated Audio Module (IAM)
10	Anti lock Brake System (ABS) control module
11	Instrument cluster
12	Touch Screen (TS)
13	Clockspring
14	Left steering wheel switchpack